9.3. Infinite Series.

⊛ Definition

An infinite series is defined as an infinite sum

$$\sum_{k=1}^{\infty} u_k = u_1 + u_2 + u_3 + \cdots + u_k + \cdots$$

⊛ Sequence → Series

Suppose we have a sequence $\{u_k\}_{k=1}^{\infty}$

Let $S_n = u_1 + u_2 + \cdots + u_n$

i.e.
$$S_1 = u_1$$
$$S_2 = u_1 + u_2$$
$$S_3 = u_1 + u_2 + u_3$$
$$\vdots$$

S_n is a sequence and is called the

<u>partial sum</u> of $\{u_n\}$.

infinite series

$$S_\infty = u_1 + u_2 + u_3 + \cdots + u_k + \cdots$$

$$= \lim_{n\to\infty} S_n$$

⊛ Geometric Sequence and Geometric Series.

⊕ Geometric Sequence:

$$a_n = a \cdot r^{n-1}$$ where a, r are constants.

⊕ Geometric Series

$$\sum_{n=1}^{\infty} a_n = a_1 + \cdots + a_n + \cdots$$

$$= a + ar + ar^2 + \cdots + ar^n + \cdots$$

Example:

① $a = 1, \quad r = 1$

$$a_n = a \cdot r^{n-1} = 1 \cdot 1^{n-1} = 1$$

$$\sum_{n=1}^{\infty} a_n = 1 + 1 + 1 + \cdots \rightarrow \text{diverse}$$

② $a = 1, \quad r = 2$

$$\sum_{n=1}^{\infty} a_n = 1 + 1 \cdot 2 + 1 \cdot 2^2 + 1 \cdot 2^3 + 1 \cdot 2^4 \cdots$$

$$= 1 + 2 + 2^2 + 2^3 + 2^4 + \cdots$$

(*) $a = 1, \quad r = \frac{1}{2}$

$$\sum_{n=1}^{\infty} a_n = 1 + 1 \cdot \frac{1}{2} + 1 \cdot \left(\frac{1}{2}\right)^2 + 1 \cdot \left(\frac{1}{2}\right)^3 + \cdots$$

$$= 1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \cdots + \cdots$$

Theorem 9.3.3

A geometric series

$$\sum_{k=0}^{\infty} ar^k = a + ar + ar^2 + \cdots + ar^k + \cdots \quad (a \neq 0)$$

converges if $|r| < 1$ and diverges if $|r| \geq 1$. If the series converges, then the sum is

$$\sum_{k=0}^{\infty} ar^k = \frac{a}{1-r}$$

OR :

$$\sum_{k=1}^{\infty} a \cdot r^k = \frac{a}{1-r} - a = a\left(\frac{1}{1-r} - 1\right)$$

$$= \frac{a \cdot r}{1-r}$$

$$\boxed{\sum_{k=1}^{\infty} a \cdot r^k = \frac{ar}{1-r}}$$

Example: Calculate

$$\sum_{n=1}^{\infty} \left(\frac{1}{2}\right)^n = \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \cdots$$

$$a = 1 \quad , \quad r = \frac{1}{2}$$

$$\Rightarrow \sum_{n=1}^{\infty} 1 \cdot \left(\frac{1}{2}\right)^n = \frac{a \cdot r}{1 - r} = \frac{1/2}{1 - 1/2} = 1$$

(*) $$\sum_{k=0}^{\infty} \frac{7}{3^k}$$ $\quad \underline{a \cdot r^k}$

$a = 7$

$$= \sum_{k=0}^{\infty} 7 \cdot \left(\frac{1}{3}\right)^k$$

$r = \frac{1}{3}$

$$= \frac{a}{1 - r} = \frac{7}{1 - 1/3} = \frac{21}{2}$$

Notice:

$$1^x = 1$$

$$(-1)^x = (-1)^x$$

$$\left(\frac{a}{b}\right)^x = \frac{a^x}{b^x}$$

$$a^{x-h} = \frac{a^x}{a^h}$$

$$a^{x+h} = a^x \cdot a^h$$

$$\left(a^x\right)^h = a^{x \cdot h}$$